

Causal Machine Learning for Sustainable Crop Yield

By

Waseem Abbas

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- Objective:

“Estimating Heterogeneous Treatment Effects of Smart Irrigation Using Double/Debiased ML”

- **Research Question:** *What is the causal effect of drip irrigation on crop yield, and for which farms does it work best?*
- **Approach:** Double/Debiased Machine Learning (DML) with Causal Forest
- **Dataset:** Smart Farming Sensor Data (500 farms, 22 features)

Kaggle Data set